Functional and differential equations – 1

Functional equations: the toolkit

Homework

13 сентября 2026 года

Theoretical minimum. The working moves are always the same: substitute special values, substitute $f(x)$ for a variable, iterate, symmetrise in x and y, and extract injectivity or surjectivity — injectivity is what lets one cancel f. Any equation that looks additive or multiplicative should be pushed towards the Cauchy equation, which is solved completely on $\mathbb{Q}$ and needs one drop of regularity over $\mathbb{R}$: continuity at a point, monotonicity, boundedness on a set of positive measure. Where the unknown appears under an integral, differentiate and gain a derivative; where it appears at shifted arguments, iterate the shift. A growth estimate combined with monotonicity is the standard engine of a non-existence proof, and a fixed point of the equation is the standard source of special values.

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1. Find all functions $f\colon \mathbb{Q} \to \mathbb{Q}$ with $f(x+y) = f(x) + f(y)$ and $f(xy) = f(x)f(y)$, and all $f\colon \mathbb{R} \to \mathbb{R}$ with the same two properties, without assuming continuity.

2. (IMC 2025 shortlist) Find all differentiable functions $f\colon \mathbb{R} \to \mathbb{R}$ such that $f(x+y) \geqslant 3^x f(y) + 2^y f(x)$ for all x, y, with $f(0) = 0$ and $f(1) = 1$.

3. Find all $f\colon \mathbb{R} \to \mathbb{R}$ with $f(f(x)) = x^2 - 2$ for all x, or prove that none exists; then decide the same question for $f(f(x)) = x^2 + 1$.

4. Find all continuous $f\colon [0,1] \to [0,\infty)$ such that $\int_0^1 f \cdot \int_0^1 f^2 \cdots \int_0^1 f^{2020} = \left(\int_0^1 f^{2021}\right)^{1010}$.

5. Find all $f\colon \mathbb{N} \to \mathbb{N}$ that are strictly increasing with $f(f(n)) = 3n$, and prove that the answer is unique; then investigate $f(f(n)) = 2n$.

6. Prove that an additive function whose graph is not dense in the plane is linear, and deduce that a Lebesgue measurable additive function is linear.

7. Find all $f\colon \mathbb{R} \to \mathbb{R}$ with $f(x+y) + f(x-y) = 2f(x)\cos y$ for all x, y, and explain why no regularity assumption is needed here.

8. Find all $f\colon \mathbb{R} \to \mathbb{R}$ satisfying $f(x)f(y) - f(xy) = x + y$ for all x, y, and all $f\colon \mathbb{R} \to \mathbb{R}$ with $f(xy) = xf(y) + yf(x)$.

9. Let $f\colon \mathbb{R} \to \mathbb{R}$ be bounded above on every bounded interval and satisfy $f(x+y) \leqslant f(x) + f(y)$ (subadditivity). Prove that $\lim_{x\to\infty} \frac{f(x)}{x}$ exists in $[-\infty, \infty)$ and equals $\inf_{x>0} \frac{f(x)}{x}$, and compare with Fekete's lemma for sequences (Limits – 1).

10. Find all pairs of functions $f, g\colon \mathbb{R} \to \mathbb{R}$ with $f(x) - f(y) = g\big(\frac{x+y}{2}\big)(x-y)$ for all x, y, assuming only that f is continuous.